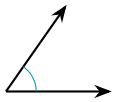
Angles — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

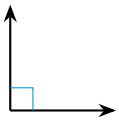
Name: _____ Date: _____

☞ *Video: Four kinds of angles*

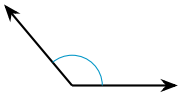
Key Idea



Acute
less than $90°$



Right
exactly $90°$



Obtuse
between $90°$ and $180°$



Straight
exactly $180°$

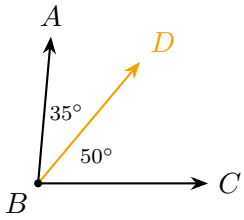
☞ *Video: Reading an angle's name*

Key Idea

In $\angle ABD$, the vertex is the **middle** letter: B. $\angle ABD$ and $\angle DBA$ name the **same** angle.

☞ *Video: Angle Addition Postulate*

Key Idea



D is inside $\angle ABC$. $m\angle ABD = 35°$, $m\angle DBC = 50°$.

$m\angle ABC = 35 + 50 =$ $85°$ — the small angles **add** to the big one.

$85° < 90°$, so it is **acute**.

☞ *Video: With algebra*

Watch Me — Worked Example

D is inside $\angle ABC$. $m\angle ABD = 2x + 5$, $m\angle DBC = 3x$, $m\angle ABC = 80°$. Find $m\angle DBC$.

$(2x + 5) + 3x = 80$ (part + part = whole) $\Rightarrow$ $5x + 5 = 80$ $\Rightarrow$ $x =$ 15

$m\angle DBC = 3(15) =$ $45°$ Check: $35 + 45 = 80$ ✓

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☞ *Video: Angle pairs*

Key Idea

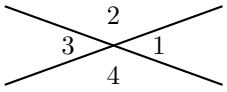
Complementary angles add to $90°$. **Supplementary** angles add to $180°$.

Memory trick: **C** comes before **S** in the alphabet — just like 90 comes before 180.

Supplement of $65°$: $180 - 65 =$ $115°$ Complement of $65°$: $90 - 65 =$ $25°$

☞ *Video: Vertical angles and linear pairs*

Key Idea



Vertical angles are across from each other and are **equal** : $\angle 1 = \angle 3$.

A **linear pair** is side by side on a line and adds to $180°$.

If $\angle 1 = 40°$: $\angle 3 =$ $40°$, $\angle 2 = \angle 4 =$ $140°$

☞ *Video: Your turn*

Your Turn

$\angle 1$ and $\angle 2$ are supplementary. $m\angle 1 = 4x + 20$, $m\angle 2 = 2x + 40$. Find x and both angles.

$(4x + 20) + (2x + 40) = 180 \quad \Rightarrow \quad 6x + 60 = 180 \quad \Rightarrow \quad x =$ 20

$m\angle 1 =$ $100°$, $m\angle 2 =$ $80°$

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